# Product Documentation

## Complete Reference Guide

*Your Name*

*Copyright © 2024 Your Company*

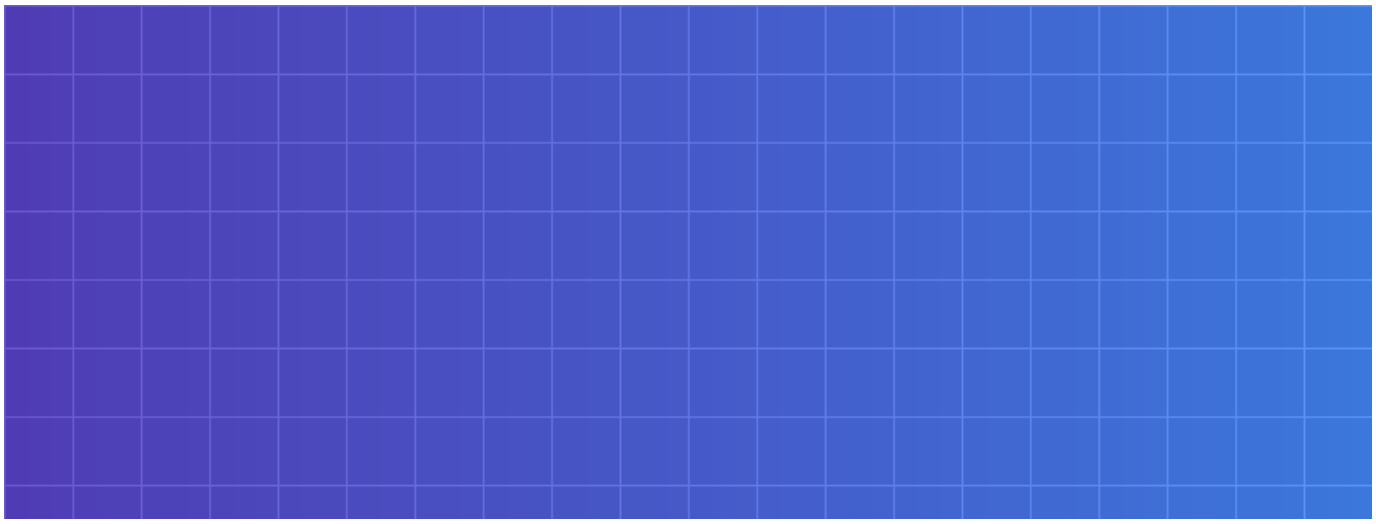
# Table of contents


1. Welcome to Product Documentation 3
   - 1.1 Quick Navigation 3
   - 1.2 What is Product? 3
   - 1.3 Key Highlights 4
2. Getting Started 5
   - 2.1 Getting Started 5
   - 2.2 Installation 8
3. Features 10
   - 3.1 Features 10
   - 3.2 Video Demo 13




Copyright © 2024 Your Company

# 1. Welcome to Product Documentation



**Product** is a powerful platform that helps teams build, ship, and scale with confidence. This documentation covers everything from initial setup to advanced configuration.

## 1.1 Quick Navigation

- **Getting Started** — Install and configure Product in minutes. Step-by-step guide to get up and running quickly.
- **Features** — Explore real-time collaboration, analytics, and integrations.
- **Video Demo** — Watch a walkthrough of key workflows and see Product in action.
- **Export to PDF** — Build the full docs as a single PDF ( `site/pdf/document.pdf` ).

## 1.2 What is Product?

Product is designed around three core principles:

**Simplicity** **Reliability** **Extensibility**

Set up in under 5 minutes. No complex configuration required — sensible defaults handle 90% of use cases out of the box.

Built on battle-tested infrastructure with 99.9% uptime SLA. Automatic failover and replication keep your data safe.

A rich plugin ecosystem and open API allow you to integrate with any tool in your stack.



Copyright © 2024 Your Company

## 1.3 Key Highlights

| Feature | Starter | Pro | Enterprise |
|---|---|---|---|
| Users | 5 | 50 | Unlimited |
| Storage | 10 GB | 100 GB | Custom |
| API Rate Limit | 1,000 req/min | 10,000 req/min | Unlimited |
| SSO / SAML | ✗ | ✓ | ✓ |
| Priority Support | ✗ | ✗ | ✓ |
| PDF Export | ✓ | ✓ | ✓ |

**New in v2.0**

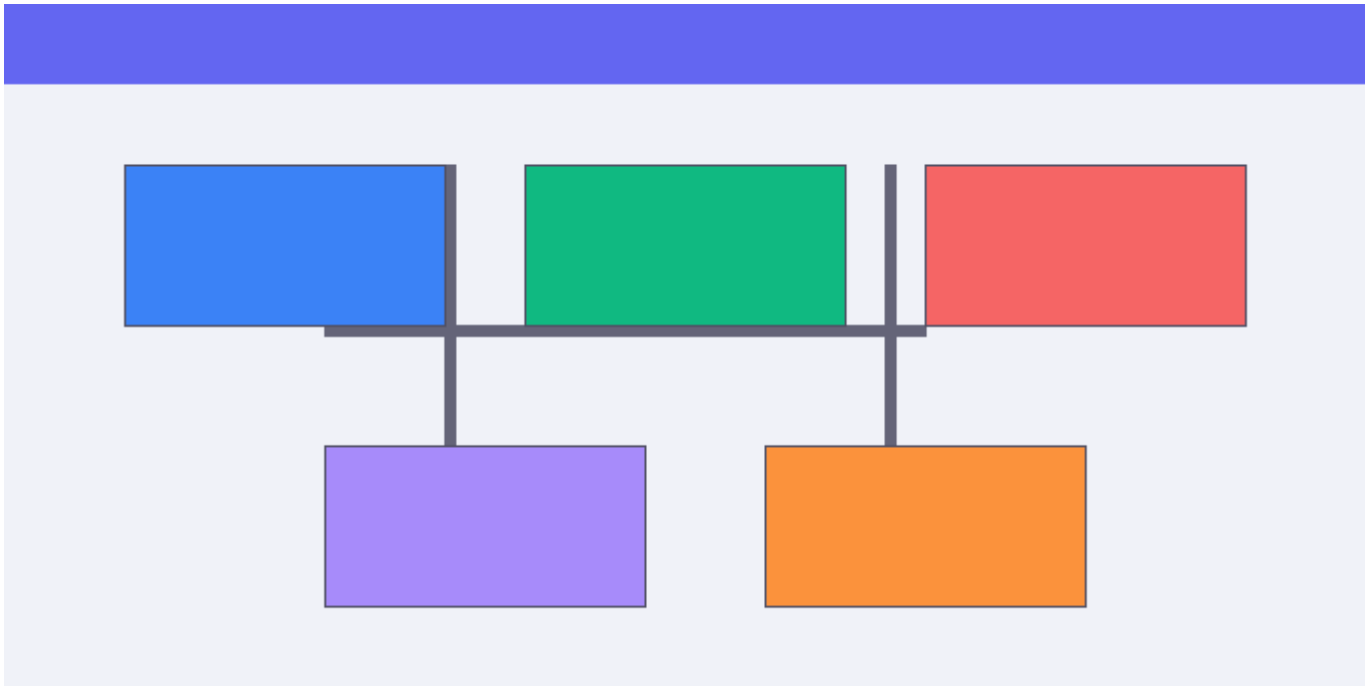
Version 2.0 introduces native PDF export, dark mode support, and a redesigned dashboard. See the changelog for full details.



Copyright © 2024 Your Company

# 2. Getting Started

## 2.1 Getting Started



This section walks you through everything you need to know to start using Product — from understanding the core concepts to configuring your first project.

### 2.1.1 Overview

Product follows a **workspace → project → resource** hierarchy. Understanding this model makes the rest of the documentation much clearer.

```
Workspace
├── Project A
│   ├── Resources
│   └── Settings
└── Project B
    ├── Resources
    └── Settings
```

#### Workspaces

A **workspace** is the top-level container for your organization. It holds all projects, members, and billing information. You typically have one workspace per company or team.

> **Workspace limits**
>
> On the Starter plan, each workspace is limited to 5 members. Upgrade to Pro or Enterprise to add more collaborators.



Copyright © 2024 Your Company

#### Projects

**Projects** are isolated environments within a workspace. Each project has its own:

- Configuration and environment variables
- Access control (invite specific members)
- Resource quotas
- Audit logs

#### Resources

**Resources** are the actual entities within a project — datasets, pipelines, models, or any custom type registered via the plugin API.

---

### 2.1.2 Core Concepts

#### Authentication

Product uses token-based authentication. Every API call requires a bearer token:

```
curl -H "Authorization: Bearer <YOUR_TOKEN>" \
     https://api.example.com/v1/projects
```

Tokens are scoped to a workspace and can be restricted to specific permissions:

| Scope | Description |
|---|---|
| `read` | Read-only access to all resources |
| `write` | Create and update resources |
| `admin` | Full control including member management |
| `billing` | Access to billing and usage data |

#### Configuration File

Every project can include a `product.yml` configuration file at its root:

```yaml
version: "2"
project:
  name: my-project
  region: us-east-1

resources:
  - name: primary-dataset
    type: dataset
    source: s3://my-bucket/data/

  - name: inference-pipeline
    type: pipeline
    steps:
      - transform
      - validate
      - export

settings:
  retention_days: 30
  notifications:
    slack_webhook: ${SLACK_WEBHOOK_URL}
```

#### Event System

Product emits events for every significant action. You can subscribe to events via webhooks or the streaming API:

```
import product_sdk as product
```


Copyright © 2024 Your Company

```python
client = product.Client(token="<YOUR_TOKEN>")

@client.on("resource.created")
def handle_created(event):
    print(f"New resource: {event.resource.name}")

@client.on("pipeline.failed")
def handle_failure(event):
    alert_team(event.error)

client.listen()
```

## 2.1.3 Architecture

The diagram below shows how components interact at runtime:

#### Data Flow

1. **Ingestion Layer** — accepts data from HTTP, message queues, or scheduled batch jobs
2. **Processing Engine** — applies transformations, validations, and enrichments
3. **Storage Layer** — persists results in columnar format with automatic indexing
4. **Query API** — exposes a SQL-compatible interface for downstream consumers

#### Fault Tolerance

Each layer implements independent retry logic with exponential backoff. If the processing engine becomes unavailable, the ingestion layer buffers incoming data in a durable queue for up to 24 hours.

> ⚠ **Data retention**
>
> Buffered data older than 24 hours is dropped by default. Configure `buffer.max_age` to increase this limit.

## 2.1.4 Next Steps

- Installation guide — install the CLI and SDK
- Features overview — explore what Product can do
- Video walkthrough — see it in action


Copyright © 2024 Your Company

## 2.2 Installation

Follow these steps to install and verify your local development environment.

### 2.2.1 Prerequisites

- **Python** 3.9 or later
- **Node.js** 18 or later (for the web dashboard)
- **Docker** 24 or later (optional, for local service emulation)

### 2.2.2 Install the CLI

macOS / Linux  Windows  Docker

```
curl -sSL https://install.example.com | sh
```

Or via Homebrew:

```
brew tap example/product
brew install product
```

Download the installer from the releases page or use `winget`:

```
winget install Example.Product
```

```
docker pull example/product-cli:latest
docker run --rm example/product-cli:latest product --version
```

Verify the installation:

```
product --version
# product v2.0.1 (darwin/arm64)
```

### 2.2.3 Install the Python SDK

```
pip install product-sdk
```

For development extras (testing utilities, type stubs):

```
pip install "product-sdk[dev]"
```

### 2.2.4 Authenticate

```
product auth login
```

This opens a browser window where you complete the OAuth flow. Your token is stored in `~/.product/credentials`.

To authenticate non-interactively (CI environments):

```
export PRODUCT_TOKEN=your_token_here
product auth verify
```

### 2.2.5 Initialize a Project

```
mkdir my-project && cd my-project
product init
```


Copyright © 2024 Your Company

You'll be prompted to choose a workspace and project name. This creates a `product.yml` in the current directory.

**You're ready**

At this point you have a working installation. Head to the Features section to start exploring what Product can do.

### 2.2.6 Building the Documentation as PDF

This documentation site is configured with the `mkdocs-with-pdf` plugin. After cloning the docs repo:

```
# Activate the virtual environment
source venv/bin/activate

# Build — PDF is generated at site/pdf/document.pdf
mkdocs build

# Or serve locally (PDF is NOT generated in serve mode)
mkdocs serve
```

The generated PDF will be at `site/pdf/document.pdf`.


Copyright © 2024 Your Company

# 3. Features

## 3.1 Features

Product ships with a broad set of built-in capabilities. This page gives you a high-level tour of each major feature area.

### 3.1.1 Real-Time Collaboration

Multiple team members can work on the same project simultaneously. Changes are broadcast over a WebSocket connection with operational-transform conflict resolution — no more "who saved last wins."

#### Presence Indicators

See exactly who is viewing or editing a resource in real time:

- **Green dot** — currently active
- **Yellow dot** — idle for more than 2 minutes
- **Gray dot** — viewing but not editing

#### Comments & Annotations

Attach threaded comments to any resource, line of configuration, or pipeline step. Comments support Markdown and can mention `@teammates` to send notifications.

> **Keyboard shortcut**
>
> Press ++cmd+shift+m++ on macOS (or ++ctrl+shift+m++ on Windows/Linux) to open the comment panel from anywhere in the UI.

---

### 3.1.2 Analytics Dashboard

The built-in analytics dashboard provides real-time and historical metrics without requiring a separate observability stack.


Copyright © 2024 Your Company

#### Available Metrics

| Metric | Description | Retention |
|---|---|---|
| Request rate | Requests per second across all endpoints | 90 days |
| Error rate | Percentage of 4xx/5xx responses | 90 days |
| P99 latency | 99th percentile response time | 30 days |
| Resource usage | CPU, memory, and storage per resource | 14 days |
| Pipeline throughput | Records processed per minute | 30 days |

#### Custom Dashboards

Create custom dashboards by combining any available metrics into a grid layout:

```yaml
# dashboard.yml
name: "Pipeline Health"
panels:
  - type: timeseries
    title: Throughput (records/min)
    metric: pipeline.throughput
    window: 1h

  - type: stat
    title: Error Rate (last 5m)
    metric: pipeline.error_rate
    threshold:
      warn: 0.01
      critical: 0.05

  - type: heatmap
    title: Latency Distribution
    metric: pipeline.latency_p99
    window: 24h
```

---

### 3.1.3 Integrations

Product connects natively with the tools your team already uses.

| Integration | Description |
|---|---|
| :simple-slack: **Slack** | Send notifications to channels or DMs when pipelines fail, succeed, or need review. |
| **GitHub** | Trigger pipeline runs on pull requests, push events, or tag creation via GitHub Actions. |
| :simple-amazonaws: **AWS** | Read from S3, write to RDS, invoke Lambda, or stream from Kinesis with zero-config auth via IAM roles. |
| **Google Cloud** | Ingest from BigQuery, Pub/Sub, and Cloud Storage. Authenticate with Workload Identity. |
| **Datadog** | Forward all Product metrics and logs directly to your Datadog account. |
| **Webhooks** | Send HTTP POST events to any endpoint for custom integrations. Includes HMAC signature verification. |

---

### 3.1.4 Access Control

Role-based access control (RBAC) is available on all plans.

#### Built-in Roles

```
Owner
├── Admin
│   ├── Editor
```


Copyright © 2024 Your Company

```
│   │   └── Viewer
│   └── Billing Manager
└── API Service Account
```

#### Custom Roles (Enterprise)

Define granular permissions with a policy language:

```
{
  "role": "pipeline-operator",
  "permissions": [
    { "resource": "pipeline", "actions": ["read", "run", "stop"] },
    { "resource": "log",      "actions": ["read"] },
    { "resource": "dataset",  "actions": ["read"] }
  ]
}
```

> ⚠️ **Permission inheritance**
>
> Permissions are additive — a user with both `Viewer` and `pipeline-operator` roles has the union of both sets of permissions.

---

### 3.1.5 Data Export

Export any dataset or pipeline output in multiple formats:

**CSV** **Parquet** **JSON Lines** **PDF Report**

```
product export my-dataset --format csv --output ./data.csv

product export my-dataset --format parquet --output ./data.parquet

product export my-dataset --format jsonl --output ./data.jsonl

# Export a formatted PDF report of a resource's summary page
product report my-dataset --format pdf --output ./report.pdf
```

---

### 3.1.6 Next Steps

- Video Demo — watch these features in action
- Installation — set up your environment


Copyright © 2024 Your Company

## 3.2 Video Demo

Watch these walkthroughs to see Product in action. All recordings are available locally in the `docs/assets/videos/` directory.

> **Adding your own videos**
>
> Place `.mp4` files in `docs/assets/videos/` and reference them with the embed code shown below. Videos are served as static assets — no external hosting required.

---

### 3.2.1 Quickstart Walkthrough

This 3-minute overview covers creating your first workspace, running a pipeline, and inspecting the results.

Your browser does not support HTML5 video. Download the video instead.

**What you'll see:**

1. Logging in and creating a workspace
2. Importing a sample dataset from S3
3. Running the built-in validation pipeline
4. Reviewing results in the analytics dashboard

---

### 3.2.2 Pipeline Configuration Deep-Dive

A 6-minute walkthrough covering advanced pipeline configuration, including multi-step transforms, conditional branching, and error handling strategies.

Your browser does not support HTML5 video. Download the video instead.

**Topics covered:**

- Writing multi-step pipeline YAML
- Using environment variable interpolation
- Configuring retries and dead-letter queues
- Connecting a Slack webhook for failure alerts

---

### 3.2.3 How to Embed a Local Video

Copy this snippet into any Markdown page and adjust the `src` path:

```
<video
  controls
  width="100%"
  style="border-radius: 8px; max-width: 800px; display: block; margin: 1rem auto;"
  poster="path/to/thumbnail.png"
>
  <source src="path/to/your-video.mp4" type="video/mp4">
  Your browser does not support HTML5 video.
</video>
```


Copyright © 2024 Your Company

**Path Reference Guide**

| Page location | Video in `assets/videos/` | Correct `src` |
|---|---|---|
| `docs/index.md` | `quickstart.mp4` | `assets/videos/quickstart.mp4` |
| `docs/features/video-demo.md` | `quickstart.mp4` | `../assets/videos/quickstart.mp4` |
| `docs/getting-started/index.md` | `quickstart.mp4` | `../assets/videos/quickstart.mp4` |

**Thumbnail / poster frame**

The `poster` attribute sets the image shown before the user hits play. Point it at any PNG or JPG in your assets folder.

---

## 3.2.4 Supported Formats

| Format | Extension | Notes |
|---|---|---|
| MPEG-4 | `.mp4` | Best compatibility — recommended |
| WebM | `.webm` | Smaller file size, modern browsers only |
| OGG | `.ogv` | Legacy fallback |

For maximum browser compatibility, provide both MP4 and WebM sources:

```
<video controls width="100%">
  <source src="../assets/videos/demo.webm" type="video/webm">
  <source src="../assets/videos/demo.mp4"  type="video/mp4">
  Your browser does not support HTML5 video.
</video>
```

---

**PDF export and videos**

When this page is exported to PDF via `mkdocs build`, video embeds are replaced by the poster image with a caption noting the original file path. This is handled automatically by the `mkdocs-with-pdf` plugin.



Copyright © 2024 Your Company